Degenerative disk disease: assessment of changes in vertebral body marrow with MR imaging.

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Abstract

The authors reviewed magnetic resonance (MR) images of 474 consecutive patients referred for lumbar spine MR imaging. Type 1 changes (decreased signal intensity on T1-weighted spin-echo images and increased signal intensity on T2-weighted images) were identified in 20 patients (4%) and type 2 (increased signal intensity on T1-weighted images and isointense or slightly increased signal intensity on T2-weighted images) in 77 patients (16%). In all cases there was evidence of associated degenerative disk disease at the level of involvement. Histopathologic sections in three cases of type 1 change demonstrated disruption and fissuring of the end plates and vascularized fibrous tissue, while in three cases of type 2 change they demonstrated yellow marrow replacement. In addition, 16 patients with end-plate changes documented with MR were studied longitudinally. Type 1 changes in five of six patients converted to a type 2 pattern in 14 months to 3 years. Type 2 changes in ten patients remained stable over a 2-3-year period. These signal intensity changes appear to reflect a spectrum of vertebral body marrow changes associated with degenerative disk disease.